EUCLID · BOOK I · PROPOSITION 5

Cross the isosceles triangle’s bridge

Equal sides give equal base angles. Euclid reaches them through two pairs of overlapping triangles and an angle subtraction, proving the exterior angles equal along the way.

THE PROPOSITION

GREEK Ἑλληνικά

Τῶν ἰσοσκελῶν τριγώνων αἱ πρὸς τῇ βάσει γωνίαι ἴσαι ἀλλήλαις εἰσίν, καὶ προσεκβληθεισῶν τῶν ἴσων εὐθειῶν αἱ ὑπὸ τὴν βάσιν γωνίαι ἴσαι ἀλλήλαις ἔσονται.

LATIN Latine

In triangulis aequicruriis anguli ad basim positi inter se aequales sunt, et productis rectis aequalibus anguli sub basi positi inter se aequales erunt.

ENGLISH reading translation

In isosceles triangles the angles at the base are equal to one another; when the equal sides are extended, the angles below the base will also be equal to one another.

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1 Begin with equal sides and extend them

GREEK Ἑλληνικά

Ἔστω τρίγωνον ἰσοσκελὲς τὸ ΑΒΓ ἴσην ἔχον τὴν ΑΒ πλευρὰν τῇ ΑΓ πλευρᾷ, καὶ προσεκβεβλήσθωσαν ἐπ’ εὐθείας ταῖς ΑΒ, ΑΓ εὐθεῖαι αἱ ΒΔ, ΓΕ·

LATIN Latine

Sit triangulus aequicrurius ΑΒΓ habens ΑΒ = ΑΓ, et producantur ΑΒ, ΑΓ in directum, ut fiant ΒΔ, ΓΕ.

ENGLISH reading translation

Let ΑΒΓ be an isosceles triangle with side AB equal to AΓ. Let BΔ and ΓΕ continue AB and AΓ in straight lines.

2 There are two angle claims

GREEK Ἑλληνικά

λέγω, ὅτι ἡ μὲν ὑπὸ ΑΒΓ γωνία τῇ ὑπὸ ΑΓΒ ἴση ἐστίν, ἡ δὲ ὑπὸ ΓΒΔ τῇ ὑπὸ ΒΓΕ.

LATIN Latine

dico, esse ∠ ΑΒΓ = ΑΓΒ et ∠ ΓΒΔ = ΒΓΕ.

ENGLISH reading translation

I say that angle ΑΒΓ equals ΑΓΒ, and angle ΓΒΔ equals ΒΓΕ.

3 Choose Z, cut AH and join the cross-lines

GREEK Ἑλληνικά

εἰλήφθω γὰρ ἐπὶ τῆς ΒΔ τυχὸν σημεῖον τὸ Ζ, καὶ ἀφῃρήσθω ἀπὸ τῆς μείζονος τῆς ΑΕ τῇ ἐλάσσονι τῇ ΑΖ ἴση ἡ ΑΗ, καὶ ἐπεζεύχθωσαν αἱ ΖΓ, ΗΒ εὐθεῖαι.

LATIN Latine

Sumatur enim in ΒΔ quoduis punctum Ζ, et a maiore ΑΕ minori ΑΖ aequalis abscindatur ΑΗ, et ducantur ΖΓ, ΗΒ rectae.

ENGLISH reading translation

Take any point Z on ΒΔ. From ΑΕ, the greater, cut off ΑΗ equal to the lesser ΑΖ, and join ΖΓ and ΗΒ.

4 Match the two larger triangles

GREEK Ἑλληνικά

ἐπεὶ οὖν ἴση ἐστὶν ἡ μὲν ΑΖ τῇ ΑΗ ἡ δὲ ΑΒ τῇ ΑΓ, δύο δὴ αἱ ΖΑ, ΑΓ δυσὶ ταῖς ΗΑ, ΑΒ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ γωνίαν κοινὴν περιέχουσι τὴν ὑπὸ ΖΑΗ·

LATIN Latine

iam quoniam ΑΖ = ΑΗ et ΑΒ = ΑΓ, duae rectae ΖΑ, ΑΓ duabus ΗΑ, ΑΒ aequales sunt altera alteri; et angulum communem comprehendunt ΖΑΗ.

ENGLISH reading translation

Since AZ equals AH and AB equals AΓ, the pairs ZA, AΓ and HA, AB are respectively equal. They contain the common angle ZAH.

5 Use I.4 for the larger pair

GREEK Ἑλληνικά

βάσις ἄρα ἡ ΖΓ βάσει τῇ ΗΒ ἴση ἐστίν, καὶ τὸ ΑΖΓ τρίγωνον τῷ ΑΗΒ τριγώνῳ ἴσον ἔσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται ἑκατέρα ἑκατέρᾳ, ὑφ’ ἃς αἱ ἴσαι πλευραὶ ὑποτείνουσιν, ἡ μὲν ὑπὸ ΑΓΖ τῇ ὑπὸ ΑΒΗ, ἡ δὲ ὑπὸ ΑΖΓ τῇ ὑπὸ ΑΗΒ.

LATIN Latine

itaque ΖΓ = ΗΒ et △ ΑΖΓ = ΑΗΒ, et reliqui anguli reliquis aequales erunt alter alteri, sub quibus aequalia latera subtendunt, ∠ ΑΓΖ = ΑΒΗ et ∠ ΑΖΓ = ΑΗΒ.

ENGLISH reading translation

Therefore ΖΓ equals ΗΒ, and triangle ΑΖΓ equals ΑΗΒ. The remaining corresponding angles are equal: ΑΓΖ to ΑΒΗ, and ΑΖΓ to ΑΗΒ.

6 Subtract the equal upper lengths

GREEK Ἑλληνικά

καὶ ἐπεὶ ὅλη ἡ ΑΖ ὅλῃ τῇ ΑΗ ἐστιν ἴση, ὧν ἡ ΑΒ τῇ ΑΓ ἐστιν ἴση, λοιπὴ ἄρα ἡ ΒΖ λοιπῇ τῇ ΓΗ ἐστιν ἴση.

LATIN Latine

et quoniam ΑΖ = ΑΗ, quarum partes ΑΒ, ΑΓ aequales, erit ΒΖ = ΓΗ.

ENGLISH reading translation

Since the whole ΑΖ equals the whole ΑΗ, and the parts ΑΒ and ΑΓ are equal, the remaining ΒΖ equals the remaining ΓΗ.

7 Match the two lower triangles

GREEK Ἑλληνικά

ἐδείχθη δὲ καὶ ἡ ΖΓ τῇ ΗΒ ἴση· δύο δὴ αἱ ΒΖ, ΖΓ δυσὶ ταῖς ΓΗ, ΗΒ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ γωνία ἡ ὑπὸ ΒΖΓ γωνίᾳ τῇ ὑπὸ ΓΗΒ ἴση, καὶ βάσις αὐτῶν κοινὴ ἡ ΒΓ·

LATIN Latine

sed demonstratum est etiam ΖΓ = ΗΒ. itaque duae rectae ΒΖ, ΖΓ duabus ΓΗ, ΗΒ aequales sunt altera alteri; et ∠ ΒΖΓ = ΓΗΒ et basis eorum communis ΒΓ.

ENGLISH reading translation

ΖΓ was also proved equal to ΗΒ. Thus ΒΖ, ΖΓ equal ΓΗ, ΗΒ respectively, angle ΒΖΓ equals ΓΗΒ, and their base ΒΓ is common.

8 Use I.4 again below the base

GREEK Ἑλληνικά

καὶ τὸ ΒΖΓ ἄρα τρίγωνον τῷ ΓΗΒ τριγώνῳ ἴσον ἔσται, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι ἔσονται ἑκατέρα ἑκατέρᾳ, ὑφ’ ἃς αἱ ἴσαι πλευραὶ ὑποτείνουσιν· ἴση ἄρα ἐστὶν ἡ μὲν ὑπὸ ΖΒΓ τῇ ὑπὸ ΗΓΒ ἡ δὲ ὑπὸ ΒΓΖ τῇ ὑπὸ ΓΒΗ.

LATIN Latine

itaque etiam △ ΒΖΓ = ΓΗΒ, et reliqui anguli reliquis aequales erunt alter alteri, sub quibus aequalia latera subtendunt. itaque ∠ ΖΒΓ = ΗΓΒ et ΒΓΖ = ΓΒΗ.

ENGLISH reading translation

The triangles ΒΖΓ and ΓΗΒ are therefore equal, with their remaining corresponding angles equal. Hence ΖΒΓ equals ΗΓΒ, and ΒΓΖ equals ΓΒΗ.

9 Subtract equal angles to expose the base angles

GREEK Ἑλληνικά

ἐπεὶ οὖν ὅλη ἡ ὑπὸ ΑΒΗ γωνία ὅλῃ τῇ ὑπὸ ΑΓΖ γωνίᾳ ἐδείχθη ἴση, ὧν ἡ ὑπὸ ΓΒΗ τῇ ὑπὸ ΒΓΖ ἴση, λοιπὴ ἄρα ἡ ὑπὸ ΑΒΓ λοιπῇ τῇ ὑπὸ ΑΓΒ ἐστιν ἴση· καί εἰσι πρὸς τῇ βάσει τοῦ ΑΒΓ τριγώνου.

LATIN Latine

iam quoniam ∠ ΑΒΗ = ΑΓΖ, ut demonstratum est, quorum partes ΓΒΗ, ΒΓΖ aequales, erit ∠ ΑΒΓ = ΑΓΒ. et sunt ad basim positi trianguli ΑΒΓ.

ENGLISH reading translation

The whole angles ABH and AΓZ were proved equal, and their parts ΓBH and BΓZ are equal. The remaining angles ABΓ and AΓB are therefore equal; these are at the base of triangle ABΓ.

10 Recover the angles below the base

GREEK Ἑλληνικά

ἐδείχθη δὲ καὶ ἡ ὑπὸ ΖΒΓ τῇ ὑπὸ ΗΓΒ ἴση· καί εἰσιν ὑπὸ τὴν βάσιν.

LATIN Latine

uerum etiam demonstratum est ∠ ΖΒΓ = ΗΓΒ; et sub basi sunt.

ENGLISH reading translation

Angle ZBΓ was also proved equal to HΓB; these are below the base.

11 Both angle claims are established

GREEK Ἑλληνικά

Τῶν ἄρα ἰσοσκελῶν τριγώνων αἱ πρὸς τῇ βάσει γωνίαι ἴσαι ἀλλήλαις εἰσίν, καὶ προσεκβληθεισῶν τῶν ἴσων εὐθειῶν αἱ ὑπὸ τὴν βάσιν γωνίαι ἴσαι ἀλλήλαις ἔσονται· ὅπερ ἔδει δεῖξαι.

LATIN Latine

Ergo in triangulis aequicruriis anguli ad basim positi inter se aequales sunt, et productis rectis aequalibus anguli sub basi positi inter se aequales erunt; quod erat demonstrandum.

ENGLISH reading translation

Therefore an isosceles triangle has equal angles at its base, and extending its equal sides gives equal angles below the base. Which was to be shown.

Stoicheia · Euclid, Elements I.1–10. Greek and Latin: J. L. Heiberg, Euclidis opera omnia, vol. I (Leipzig, 1883), pp. 4–31; public domain. Latin is Heiberg’s nineteenth-century translation. English reading and notes: Stoicheia, 2026, MIT. Source and edition notes: antonsoo.github.io/stoicheia/